

Car Accident Severity Prediction

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1 Introduction

Road traffic accidents represent a major public safety concern worldwide, leading to injuries, fatalities, and significant economic losses. Predicting the severity of car accidents can help authorities and policymakers design better safety measures and allocate emergency resources more effectively. In this project, machine learning techniques are applied to predict accident severity based on historical accident data.

The goal of this study is to build and evaluate a machine learning model capable of classifying accident severity into different levels using relevant environmental and road-related features. The project follows a standard machine learning pipeline, including data preprocessing, model training, and evaluation.

2 Dataset Description

The dataset used in this project contains historical road accident records from the United Kingdom. It includes information related to accident conditions, such as weather, lighting, road surface, number of vehicles involved, and speed limits. The original dataset consists of more than 1.5 million records with multiple attributes.

Due to computational constraints, a random sample of 100,000 records was selected for analysis. After removing missing values and selecting the most relevant features, the final dataset was used to train and test the machine learning model. The target variable, accident severity, was encoded into numerical labels for model training.

3 Methodology

3.1 Data Preprocessing

Data preprocessing involved selecting relevant features, handling missing values, encoding categorical variables using one-hot encoding, and scaling numerical features where necessary. The dataset was split into training and testing sets using an 80/20 ratio to evaluate model performance on unseen data.

3.2 Model Selection

A Random Forest classifier was selected as the primary model due to its robustness, ability to handle non-linear relationships, and strong performance on tabular data. Support Vector Machines (SVM) and Multi-Layer Perceptron (MLP) neural networks were also considered and discussed theoretically, as they are commonly used for classification problems.

4 Results and Evaluation

The Random Forest model achieved strong performance on the test dataset. The evaluation metrics used include accuracy, F1-score, confusion matrix, and classification report. The model achieved an accuracy of approximately **85%** and an F1-score of **78%**, indicating good overall classification performance.

The confusion matrix revealed that the model performs well on majority classes but shows reduced performance for minority classes, particularly fatal accidents. This behavior is expected due to class imbalance in the dataset, where severe accidents occur less frequently than minor ones.

5 Ethical Considerations

The use of machine learning for car accident severity prediction involves several important ethical considerations. Although such models can support decision-making and improve road safety planning, they must be developed and deployed responsibly. The dataset used in this study consists of historical accident records and does not include personally identifiable information, which helps reduce privacy concerns. However, careful handling of data is still required to prevent misuse or unintended disclosure.

Another key ethical issue is data bias and class imbalance. Severe and fatal accidents occur far less frequently than minor accidents, leading to imbalanced datasets. As a result, predictive models may perform well on majority classes while underperforming on minority classes, such as fatal accidents. This imbalance can limit the model’s reliability for critical safety decisions and highlights the importance of transparent reporting and careful interpretation of results.

Model interpretability and fairness are also essential considerations. Complex models such as Random Forests, Support Vector Machines, and Neural Networks can act as “black boxes,” making it difficult to explain individual predictions. In safety-related applications, lack of interpretability may reduce trust and accountability. Therefore, such models should be used as decision-support tools rather than as fully autonomous systems.

Finally, ethical responsibility requires acknowledging the limitations of the model and avoiding overreliance on automated predictions. Accident severity prediction systems should complement human expertise and be continuously monitored and updated. Proper validation, fairness analysis, and responsible deployment are crucial to ensure that these models contribute positively to public safety without causing unintended harm.

6 Conclusion

In this project, a machine learning-based approach was used to predict car accident severity using historical accident data. The Random Forest model demonstrated strong predictive performance and proved suitable for this classification task. Despite limitations related to class imbalance and computational constraints, the results highlight the potential of machine learning techniques in supporting road safety analysis. Future work may focus on handling class imbalance more effectively and exploring advanced models with improved interpretability.